

Review

A comprehensive survey on secure healthcare data processing with homomorphic encryption: attacks and defenses

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Abstract

Healthcare data has risen as a top target for cyberattacks due to the rich amount of sensitive patient information. This negatively affects the potential of advanced analytics and collaborative research in healthcare. Homomorphic encryption (HE) has emerged as a promising technology for securing sensitive healthcare data while enabling computations on encrypted information. This paper conducts a background survey of HE and its various types. It discusses Partially Homomorphic Encryption (PHE), Somewhat Homomorphic Encryption (SHE), Fully Homomorphic Encryption (FHE) and Fully Leveled Homomorphic Encryption (FLHE). A critical analysis of these encryption paradigms' theoretical foundations, implementation schemes, and practical applications in healthcare contexts is presented. The survey encompasses diverse healthcare domains. It demonstrates HE's versatility in securing electronic health records (EHRs), enabling privacy-preserving genomic data analysis, protecting medical imaging, facilitating privacy-preserving machine learning (ML), supporting secure federated learning, ensuring confidentiality in clinical trials, and enhancing remote monitoring and telehealth services. A comprehensive examination of potential vulnerabilities in HE systems is conducted. The research systematically investigates various attack vectors, including side-channel attacks, key recovery attacks, chosen plaintext attacks (CPA), chosen ciphertext attacks (CCA), known plaintext attacks (KPA), fault injection attacks (FIA), and lattice attacks. A detailed analysis of potential defense mechanisms and mitigation strategies is provided for each identified threat. The analysis underscores the importance of HE for long-term security and sustainability in healthcare systems.

Keywords Homomorphic encryption · Healthcare security · Cryptanalysis · Federated learning · Privacy-preserving computation · Genomic data protection

1 Introduction

The digitisation of patient records and the increasing reliance on data-driven medical applications have exponentially amplified health-related data volume and sensitivity [1]. The advent of technologies such as EHRs [2–4], telemedicine [5, 6], and personalised medicine [7, 8] underscores the necessity for robust data privacy and security mechanisms. The importance of safeguarding patient information from unauthorised access and ensuring health data's confidentiality, integrity, and availability has never been more critical.

HE emerges as a revolutionary cryptographic solution poised to address these pressing concerns. HE allows computations to be performed directly on encrypted data, unlike traditional encryption data, which requires data to be

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decrypted before processing. HE computations performed on encrypted data will generate encrypted results that match the output of operations performed on plaintext. [9]. This capability balances data usability and privacy while enabling secure data processing in untrusted environments [10]. Data can be processed without compromising the confidentiality of sensitive health information in the cloud [11]. Researchers can perform statistical analyses, predictive modelling, and personal health information processing on encrypted data without decryption.

The development of HE began with the PHE concept, moving on to SHE and then FHE. PHE schemes need more flexibility and expressiveness for computations since they allow only a limited number of operations on encrypted data. This limitation led to the development of SHE, which expanded the range of permissible operations. However, it still imposed constraints on the complexity and number of computations. The breakthrough came with the concept of FHE, which enables arbitrary computations on encrypted data without any limitations. HE is emerging as an important cryptography technique in the Healthcare domain. The application of HE in healthcare aligns with the principles of sustainable development. HE can accelerate medical research by promoting secure and ethical data sharing. Improving medical research leads to the development of innovative treatments and diagnostic tools. HE improves global health outcomes and promotes well-being, aligning with the United Nations' Sustainable Development Goals.

HE has various types due to the development history and the usage of each homomorphic algorithm. They are PHE, SHE, FHE, and fully levelled homomorphic encryption. Each of these types is explored along with its corresponding cryptographic schemes. This work investigates the role of HE in healthcare and emphasises its potential to enhance data security and privacy. It demonstrates how HE safeguards sensitive medical information and facilitates secure computations on encrypted data. The discussion highlights the contribution of HE to sustainable development in healthcare. It examines the use of HE in the healthcare domain, including secure EHRs and ML in healthcare. The paper also explores known attacks and potential threats towards HE and discusses countermeasures for these risks. The analysis underscores the significance of HE for long-term security and sustainability in healthcare systems.

The remainder of this work is structured as follows: Section 2 examines various HE types. Section 3 explores the applications of HE within the healthcare sector. Section 4 addresses potential attacks on HE and possible defensive strategies against these threats. Section 5 discusses open research directions and future work. Finally, Section 6 concludes the paper, summarising the key findings and contributions.

2 Overview of homomorphic encryption

HE is a cryptographic technique that allows computations on encrypted data without decrypting it. It preserves the privacy and security of the data throughout the computation process. HE is particularly valuable in medical imaging, genomics data analysis, clinic trials, and secure remote monitoring [12–15]. HE addresses the challenge of maintaining patient privacy while allowing advanced data processing and analysis by enabling secure computations on sensitive patient data. Researchers can perform complex analyses on encrypted genomics datasets to identify genetic markers associated with diseases without exposing the raw data [16]. Other than that, HE allows for processing encrypted MRI or CT scans in medical imaging to assist in diagnosis and treatment planning while keeping the images confidential [13]. Studies have demonstrated that HE can protect patient data, enable secure collaborative research, and enhance the development of precision medicine without compromising data privacy [17, 18].

2.1 Types of homomorphic encryption

HE has been separated into four types to cater to specific computational needs and applications: PHE, SHE, FHE, and FLHE for ML. Each variant addresses distinct challenges and provides unique benefits. Table 1 shows SHE and PHE schemes, and Table 2 shows FHE schemes.

2.1.1 Partially homomorphic encryption

PHE is a form of encryption that supports either addition or multiplication. RSA [19] and Paillier [20] are classic examples of PHE schemes. RSA supports multiplication computation while Paillier supports addition computation. PHE is predominantly used in scenarios where specific operations must be performed securely without revealing the underlying data. One prominent example is the use of the Paillier cryptosystem [20] for privacy-preserving medical data analysis. Paillier cryptosystem allows computations on encrypted medical data without decrypting due to its additive homomorphic

Table 1 Summary of PHE and SHE schemes

Scheme Type	Scheme name	Year	Author(s)	Key features	Homomorphic operations	Security basis
SHE	BGN [35]	2005	Boneh, Goh, Nissim	First scheme to support both addition and one multiplication	Addition, one multiplication	Subgroup decision problem
Leveled FHE	BV [36]	2011	Brakerski, Vaikuntanathan	Introduced modulus switching technique	Leveled FHE without bootstrapping	Ring Learning With Errors (Ring-LWE)
Leveled FHE	GSW [37]	2013	Gentry, Sahai, Waters	Introduced approximate eigenvector method	Leveled FHE, faster homomorphic operations	Learning With Errors (LWE)
PHE (Additive)	Paillier [38]	1999	Pascal Paillier	Probabilistic encryption	Addition	Decisional Composite Residuosity Assumption
PHE (Additive)	Benaloh [39]	1994	Josh Benaloh	Supports efficient decryption of small plaintexts	Addition	Higher Residuosity Problem
PHE (Multiplicative)	ElGamal [40]	1985	Taher ElGamal	Probabilistic encryption	Multiplication	Decisional Diffie-Hellman Assumption
PHE (Multiplicative)	Goldwasser-Micali [41]	1984	Goldwasser, Micali	First probabilistic public-key encryption	XOR (bitwise addition mod 2)	Quadratic Residuosity Problem
PHE (Multiplicative)	Unpadded RSA [42]	1978	Rivest, Shamir, Adleman	Deterministic encryption	Multiplication	Factoring Problem

Table 2 Summary of FHE schemes

Name	Year	Author(s)	Base	Type	Key algorithms	Base scheme
Gentry's Scheme [23]	2009	Craig Gentry	Lattice-based	Bootstrapping	KeyGen, Encrypt, Decrypt, Add, Mult, Decrypt	Ideal lattices
DGHV [43]	2010	van Dijk, Gentry, Halevi, Vaikuntanathan	Integer-based	Approximate GCD	KeyGen, Encrypt, Decrypt, Evaluate	Gentry's scheme
BGV [44]	2011	Brakerski, Gentry, Vaikuntanathan	LWE-based	Modulus switching	KeyGen, Encrypt, Decrypt, Add, Mult, Refresh	Regev's LWE
FV (Fan-Vercauteren) [45]	2012	Fan, Vercauteren	Ring-LWE-based	Scale-invariant	KeyGen, Encrypt, Decrypt, Add, Mult	Regev's LWE, BGV
GSW [37]	2013	Gentry, Sahai, Waters	LWE-based	Approximate eigenvectors	KeyGen, Encrypt, Decrypt, Eval	Regev's LWE
TFHE [46]	2016	Chillotti, Gama, Georgieva, Izabachène	LWE/Ring-LWE-based	Bootstrapping-friendly	KeyGen, Encrypt, Decrypt, Bootstrap	GSW
CKKS (BFV) [47]	2017	Cheon, Kim, Kim, Song	Ring-LWE-based	Approximate arithmetic	KeyGen, Encrypt, Decrypt, Add, Mult, Rescale	BGV, FV
FHEW [48]	2014	Ducas, Miccancio	LWE-based	Fast bootstrapping	KeyGen, Encrypt, Decrypt, Bootstrap	Regev's LWE, GSW
HElib [49]	2013	Halevi, Shoup, et al.	BGV-based	Optimized implementation	Various (implements BGV, CKKS)	BGV

property. Action such as calculating aggregate statistics from patient records can be performed. This capability is crucial for conducting research, analysing health trends, and developing population-level insights while upholding patient privacy and complying with data protection regulations. PHE schemes maintain lower computational overhead than SHE and FHE because they focus on addition or multiplication. This nature of PHE makes it suitable for applications with limited computational resources [21, 22].

2.1.2 Somewhat homomorphic encryption

SHE extends the capabilities of PHE by allowing a limited number of both additive and multiplicative operations on encrypted data. However, the number of operations is restricted by the scheme's noise growth. SHE will eventually corrupt the ciphertext if too many operations are performed. Craig Gentry presented one of the first SHE schemes in 2009 [23], using ideal lattices to achieve homomorphic properties and opening up possibilities for more complex computations on encrypted data. However, SHE schemes still faced limitations in the complexity and number of computations they could handle. SHE is valuable in applications where a limited set of operations needs to be performed on sensitive data [24]. SHE comes in handy for secure outsourcing of computations to the cloud. Clients such as hospitals and institutions can offload their computations to the cloud without exposing their data [25]. SHE can also be used for secure data aggregation in sensor networks where data from multiple sensors is aggregated without revealing individual sensor readings [26]. The invention of SHE was driven by the need to support more complex computations on encrypted data while managing computational efficiency. SHE broadens the scope of applications beyond what PHE can handle by allowing types of operations [27].

2.1.3 Fully homomorphic encryption

FHE is the most powerful type of HE as it supports unlimited numbers of both additive and multiplicative operations on encrypted data. Rivest, Adleman, and Dertouzos first proposed the concept of FHE in 1978 [28]. A practical FHE scheme was realised with Craig Gentry's breakthrough in 2009 [23]. FHE has transformative potential across various domains, such as the healthcare and financial domains. FHE is applied to healthcare domain privacy-preserving genomics data analysis where sensitive genetic information is processed [24]. FHE enables secure computations on encrypted financial data in the financial domain and facilitates privacy-preserving audits and fraud detection [29]. Additionally, FHE is crucial for secure multi-party computation. It allows multiple parties to jointly compute a function over their inputs while keeping those inputs private [30, 31]. The primary motivation behind FHE was to enable arbitrary computation on encrypted data. It also solves the fundamental problem of performing complex data analysis while maintaining privacy. FHE addresses the limitations of PHE and SHE by providing a comprehensive solution for secure data processing, along with significant computational and performance challenges [32–34].

2.1.4 Fully leveled homomorphic encryption

FLHE is a specialised form of FHE optimised for ML applications. FLHE allows for a predetermined number of operations that align well with the layered structure of ML models. FLHE supports deep learning computations on encrypted data without the prohibitive overhead of full FHE by carefully managing the noise growth [44]. FLHE is particularly suited for privacy-preserving ML since the models are trained and evaluated on encrypted data. This has profound implications for sensitive applications such as medical diagnostics. Patient data can be securely analysed to train predictive models without compromising privacy with the use of FLHE [50, 51]. FLHE also facilitates secure federated learning and enables multiple organisations to collaboratively train a model on their combined data without exposing individual datasets [52]. The invention of FLHE was driven by the need to balance computational feasibility and security in machine-learning contexts. Traditional FHE is often too slow for practical use in deep learning. FLHE addresses this by providing a tailored solution that leverages the hierarchical nature of neural networks, optimising performance while still maintaining robust security [53].

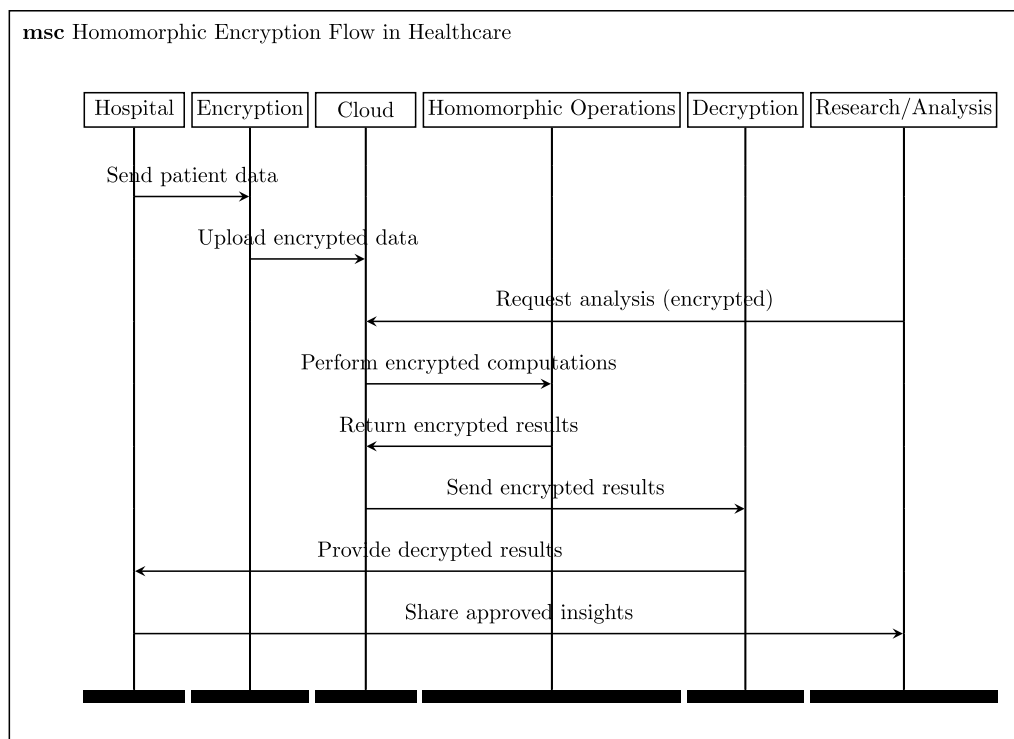


Fig. 1 Homomorphic encryption flow in healthcare

3 Application of homomorphic encryption in healthcare domain

The section discusses various applications of HE in healthcare as HE enables secure handling of sensitive information across multiple areas of healthcare. HE is becoming one of the critical solutions in EHRs, genomic data analysis, secure medical imaging, clinical trials and privacy-preserving ML. This highlights its potential to revolutionise data security and privacy in the medical field.

3.1 Secure electronic health records

EHRs are digital versions of patients' paper charts and contain comprehensive health information. It includes medical history, diagnoses, medications, immunisation dates, and laboratory test results. The sensitive nature of this information makes it a prime target for cyberattacks [54, 55]. Traditional encryption methods protect data at rest and in transit but require decryption for processing. The traditional method can expose the data to potential breaches [4]. Figure 1 demonstrates the flow of HE when it is involved in the healthcare application. HE enables secure querying and computation on EHRs without decrypting the data. A hospital can use HE to search encrypted EHRs to retrieve specific patient information or aggregate data for research purposes. This ensures that sensitive patient information remains confidential throughout the process. Additionally, HE allows for the secure sharing of EHRs between healthcare providers and researchers. It helps collaborative care and studies without compromising patient privacy [56, 57].

3.2 Privacy-preserving genomic data analysis

Genomic data analysis involves examining an individual's DNA to identify genetic variations that may influence health and disease. This data is invaluable for personalised medicine because treatments are tailored to an individual's genetic makeup in this area [15]. However, this raises significant concerns since genomic data is highly sensitive and personal. HE allows researchers and clinicians to perform complex genomic analyses on encrypted data. For instance, HE can be used to conduct genome-wide association studies on encrypted genomic datasets. It can identify genetic markers associated with specific

diseases without revealing individual genetic information. This ensures that participants' privacy is preserved while enabling groundbreaking research in genetics and personalised medicine [51].

3.3 Secure medical imaging

Medical imaging, such as MRI, CT scans, and X-rays, is a cornerstone of modern diagnostics. These images contain detailed information about a patient's internal structures and are critical for accurate diagnosis and treatment planning [58]. However, storing and sharing these images securely is challenging due to their size and sensitivity. HE can be applied to medical imaging to enable secure analysis and sharing of encrypted images. For instance, HE allows radiologists to perform image processing tasks, such as enhancing image quality or identifying abnormalities, directly on encrypted images. This means that images can be securely stored in the cloud and accessed by authorised personnel without ever exposing the raw data [59].

3.4 Privacy-preserving machine learning (ML)

ML has the potential to revolutionise healthcare by providing predictive analytics, personalised treatment recommendations, and automated diagnostics. However, training and deploying ML models on sensitive healthcare data poses significant privacy risks [60]. FLHE is particularly well-suited for privacy-preserving ML in healthcare. It enables the training and inference of ML models on encrypted data, ensuring that sensitive patient information is never exposed [51, 61]. For example, an ML model can be trained on encrypted patient records to predict disease outbreaks or identify high-risk patients without accessing the raw data. This allows healthcare providers to leverage the power of ML while maintaining strict privacy standards.

3.5 Secure federated learning

Federated learning is an approach where multiple institutions collaboratively train an ML model without sharing their raw data [62]. This is particularly valuable in healthcare, considering data is often siloed across different hospitals and research centres. HE enhances federated learning by ensuring that the data contributed by each institution remains encrypted throughout the training process [63, 64]. By combining HE and Federated learning, hospitals can collaboratively train a predictive model for disease diagnosis using their encrypted patient data. HE ensures that each hospital's data remains confidential even as the aggregated model benefits from a larger and more diverse dataset [65].

3.6 Privacy-preserving clinical trials

Clinical trials are essential for developing new treatments and drugs. However, they involve collecting and analysing vast amounts of sensitive patient data which is raising significant privacy concerns [14, 66]. To address these concerns, HE has been introduced to secure the analysis of clinical trial data. For example, researchers can perform statistical analyses on encrypted datasets to evaluate the efficacy and safety of new treatments without accessing the raw data. This ensures that participants' privacy is protected throughout the trial [67].

3.7 Secure remote monitoring and telehealth

Remote monitoring and telehealth services have become increasingly important in the wake of the COVID-19 pandemic. These services rely on continuous data collection from patients, which involves sensitive health information [68, 69]. HE enables secure remote monitoring by allowing healthcare providers to analyse data from wearable devices and home health monitors without accessing the raw data. HE can be used to securely monitor a patient's vital signs and detect anomalies that require medical attention. This ensures that patient data remains confidential while enabling timely and effective remote care [70].

4 Homomorphic encryption attacks and their defenses

Cryptography attacks are malicious attempts to compromise the security of cryptographic systems, aiming to exploit vulnerabilities and gain unauthorised access to sensitive information. Section 4 will delve into the potential attacks targeting HE systems and the corresponding defense mechanisms. Existing security concerns will be examined from three

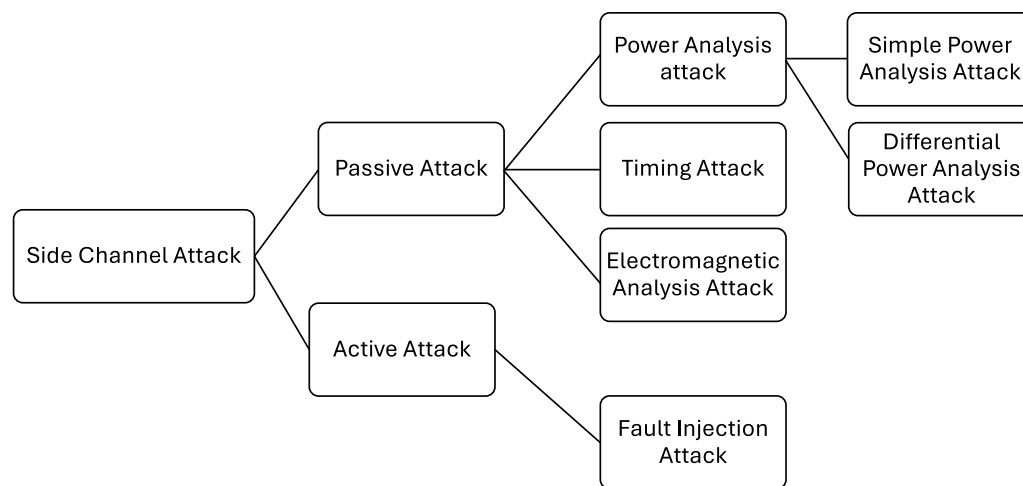


Fig. 2 Types of side channel attack

distinct perspectives. The first perspective focuses on historical attacks that have been successfully executed against HE systems. These include side-channel attacks, KPA, CPA and CCA. Each of these attack methods has demonstrated the ability to compromise the security of HE under certain circumstances. The second perspective considers scheme-based vulnerabilities, which are potential attacks that exploit weaknesses in specific HE schemes. A primary example in this category is lattice attacks, which target the mathematical structures underlying many HE systems. The third perspective addresses algorithm-based vulnerabilities, which are possible attacks that target weaknesses in the encryption algorithm itself. This category includes FIA and other forms of key recovery attacks, distinct from those in the first category. A comprehensive understanding of the challenges HE faced will be provided by analysing these three perspectives. It will help the development of more effective defense strategies to enhance the security and reliability of HE applications.

4.1 Side-channel attacks and defenses

A side-channel attack is a security exploit that aims to gather information from physical emanations such as timing, power consumption, or electromagnetic leaks. Side-channel attacks pose a significant threat to the HE system as they can extract sensitive information without directly attacking the encryption scheme itself [71–73]. Figure 2 illustrates the presence of two distinct types of side-channel attacks: passive attack and active attack. Passive side-channel attacks include power analysis attacks, timing attacks, and electromagnetic analysis attacks. Active side-channel attacks involve FIA. Passive side-channel attacks will be discussed in Sect. 4.1, while fault-injection attacks will be covered in Sect. 4.6.

4.1.1 Timing attack

A timing attack occurs when an attacker attempts to compromise a cryptosystem by analysing the time taken to execute cryptographic algorithms. It was first introduced in Kocher's work [74], explaining a timing attack occurs because every logical operation execution time varies based on the input. The input can be reverse-engineered by the attacker with precise measurements of the time for each operation. Sensitive information can be easily exposed through timing information compared to using cryptanalysis of known plaintext and ciphertext pairs [75]. Timing attacks can be combined with cryptanalysis, such as KPA, known ciphertext attacks and CCA to increase the possibility of information leakage. Information can leak from a system through measurement of the time it takes to respond to certain queries. Information such as cryptographic system design, the CPU running the system, the algorithm used, assorted implementation details, timing attack countermeasures, and the accuracy of the timing measurements are strongly dependent on defining how much the timing information gained can benefit the attacker. Any algorithm that has data-dependent timing variation can be vulnerable to timing attacks. It is difficult to remove timing dependencies in some algorithms that use low-level operations that frequently exhibit varied execution times. In the work by Cheng [76], they perform a cache-timing attack targeting the Barrett multiplication module in the SEAL library. They successfully exhibit a cache-timing vulnerability that exploits the existence of extra-reductions as side-channel leakage. In [77], the authors present a novel timing analysis

technique that estimates the error value of a final ciphertext after homomorphic gate computation. They exploit the timing channel to infer the error range, with applicability to both schemes with and without bootstrapping.

4.1.2 Power analysis attack

Power analysis is a type of side-channel attack where an attacker examines the power consumption of a cryptographic hardware device. These exploits take advantage of the device's physical design since semiconductor devices function according to the laws of physics, where small alterations in voltage demand minor movements of electric charges. By measuring these minor movements of electric charges, attackers can collect information about the processed data. [78]. Power analysis is divided into two categories: Simple Power Analysis and Differential Power Analysis (DPA). Simple Power Analysis involves the straightforward interpretation of power traces or graphs of electrical activity over time. In opposition, DPA is a more sophisticated technique, using statistical analysis of data from multiple cryptographic operations to deduce intermediate values within cryptographic computations [79, 80]. Both Simple Power Analysis and DPA were introduced to the cryptography community in 1998 by Paul Kocher, Joshua Jaffe, and Benjamin Jun [81]. Power analysis allows an attacker to gain insight into what is happening inside hardware that is otherwise considered tamper-resistant. For instance, in the DES key schedule, 28-bit key registers are rotated. Some implementations check if the least significant bit is a 1; if so, the register shifts right, and a one is prepended. If the bit is 0, the register shifts without prepending. Power analysis can detect these differences, allowing an adversary to infer the bits of the secret key. Even cryptographic algorithms like AES and triple-DES, which are mathematically robust, can be easily broken using power analysis attacks. This makes power analysis a combination of both algorithmic cryptanalysis and implementation-level vulnerabilities. This work [82] presents the first power-based side-channel attack targeting the Gaussian sampling process in the SEAL HE library, successfully extracting plaintexts by exploiting vulnerabilities in the encryption phase, with SEAL v3.2 as the primary victim. Another work [83] explores correlation power analysis (CPA) attacks on the AES-128 algorithm. It provides a theoretical overview of power analysis attacks and applies the technique to an FPGA implementation of an AES core. Using 10,000 traces, the CPA attack successfully extracted all bytes of the AES-128 final round key. Based on these findings, it is crucial to establish reliable countermeasures capable of addressing the problems raised by power analysis attacks and guaranteeing the effectiveness of HE implementations in actual applications.

4.1.3 Electromagnetic-based (EM) attack

An EM side-channel attack exploits the electromagnetic emissions from an electronic device as a source of information leakage. These attacks are non-invasive, meaning the attacker does not need physical access to the device, which makes them particularly effective and easy to execute [84]. All that is required is a near-field probe and an oscilloscope. Unlike some power-based attacks that may need specialised tools or IC decapsulation, EM attacks don't require any modifications to the targeted device. The non-intrusive nature of EM attacks also makes it nearly impossible for the victim to detect them, further enhancing their effectiveness as they are difficult to identify and prevent [85]. EM attacks have demonstrated a higher signal-to-noise ratio (SNR) compared to power-based attacks, which reduces the amount of data collection required to eliminate noise. Similar to power analysis attacks, EM attacks are categorised into two main types: Simple Electromagnetic Analysis [86] and Differential Electromagnetic Analysis [87]. The key distinction between the two is that Simple Electromagnetic Analysis involves directly interpreting the data traces, while Differential Electromagnetic Analysis collects numerous traces and applies differential statistical techniques to identify data-dependent correlations. While Differential Electromagnetic Analysis is more powerful and versatile, it is also more complex and time-intensive. With the rise of ML, analysis techniques utilising pattern recognition and classification are becoming more prevalent, making EM attacks even more effective. EM-based attacks have successfully extracted sensitive data from various cryptographic devices. For instance, Fox-IT demonstrated the strength of EM attacks by breaking an AES-256 cryptographic core within 5 min from a distance of 1 m [88]. Additionally, research has highlighted the effectiveness of EM attacks on IoT devices, further showcasing their versatility and impact.

4.1.4 Defense mechanisms 1: constant-time algorithms

Implementing algorithms in constant time helps mitigate timing attacks by ensuring that all operations take the same amount of time, regardless of the input values. This prevents attackers from gaining insights based on the time taken to perform operations on encrypted data. By eliminating timing variations, constant-time algorithms reduce the amount

of information that attackers can gather through timing analysis [89]. In the work of Almeida et al. [90], it is proven that constant time is working when it comes to defending against side-channel attacks. The paper presents a methodology for proving the security of cryptographic implementations against timing attacks, highlighting the challenges of bridging provable security and practical implementations and demonstrating a proof-of-concept on the constant-time MEE-CBC construction. Constant-time programming aims to eliminate these timing variations by ensuring that the execution time of a cryptographic operation is independent of the input values, including secret keys. One of the primary techniques in constant-time programming is to avoid conditional statements that depend on secret data. Instead of using 'if-else' constructs, constant-time code uses arithmetic operations and bitwise operations to achieve the same result without introducing timing variations. Constant-time programming ensures that memory access patterns are uniform and do not depend on secret data. This can be achieved by accessing all elements of an array or data structure in a fixed pattern, regardless of the actual data being processed. Ensuring that arithmetic operations take a constant amount of time regardless of the operands is another crucial aspect of constant-time programming. This can involve using fixed-point arithmetic or other techniques to ensure uniform execution times. Designing algorithms from the ground up to be constant-time can also be an effective strategy. This involves choosing data structures and algorithms that inherently avoid timing variations. For example, using a constant-time hash function or encryption algorithm can eliminate timing attack vectors. Authors claim that constant-time programming is a robust countermeasure against timing side-channel attacks, including those introduced by speculative execution vulnerabilities like Spectre [91].

4.1.5 Defense mechanisms 2: noise injection

Techniques such as DPA resistance involve balancing the power consumption of cryptographic devices to ensure that power usage does not vary significantly with different inputs. This reduces the effectiveness of power analysis attacks. Methods like dynamic voltage scaling, power masking, and current flattening are used to make power consumption patterns less predictable and harder to analyse [92, 93]. Researchers introduce a binary field ECC processor designed to resist DPA by using randomised Montgomery multiplication and division. The design significantly improves speed and flexibility, achieving up to 50% performance gain in Field-Programmable Gate Arrays and 36% in Application-Specific Integrated Circuit evaluations compared to earlier work. By incorporating bit-parallel architecture and module duplication, the design offers enhanced efficiency despite a slight increase in resource use [94].

4.1.6 Defense mechanisms 3: power analysis countermeasures

Power analysis countermeasures are crucial in defending against attacks that exploit the power consumption patterns of cryptographic devices. Techniques like DPA resistance aim to minimise the variations in power usage during cryptographic operations. By ensuring that the power consumption remains relatively constant, attackers find it more difficult to extract useful information. Common methods used in this defense include dynamic voltage scaling, power masking, and current flattening. These approaches work by obscuring power consumption patterns, making them less predictable and, therefore, harder to analyse. Researchers have also developed specific hardware solutions, such as a bit-parallel elliptic curve cryptography (ECC) coprocessor, which is designed to resist differential power analysis attacks effectively [94]. This coprocessor operates by balancing power consumption across operations, reducing the risk of successful attacks.

4.1.7 Defense mechanisms 4: shielding and isolation

Shielding and isolation techniques offer physical protection for cryptographic devices against side-channel attacks. Shielding involves encasing devices in materials that block electromagnetic emissions. This prevents attackers from intercepting signals that could reveal sensitive information. Another layer of defense comes from isolating critical components. By separating these components from other parts of the system, the leakage of information through unintended channels is minimised. Additionally, tamper-resistant hardware further strengthens this defense. When combined, shielding and isolation methods significantly reduce the risk of information being extracted through electromagnetic or other side-channel means [95]. These techniques are especially important in environments where physical access to cryptographic devices is possible.

4.2 Key recovery attacks and defenses

A key recovery attack is a type of cryptographic attack where the attacker aims to retrieve the secret decryption key used in a cryptographic algorithm. By obtaining the key, the attacker can decrypt any ciphertext encrypted with that key, effectively breaking the confidentiality of the encrypted data. Key recovery attacks can target various cryptographic systems, including symmetric and asymmetric encryption schemes, and they exploit weaknesses in the algorithm, implementation flaws, or inadequate key management practices [96].

Performing a key recovery attack involves several steps. The attacker may use various techniques, such as brute force, where all possible keys are systematically tried until the correct one is found, or more sophisticated methods, like side-channel attacks, which exploit physical leakages from the cryptographic device, such as timing information, power consumption, or electromagnetic emissions. Cryptanalysis techniques, such as linear and differential cryptanalysis, can also be employed to reduce the complexity of finding the key by analysing the relationship between plaintexts and ciphertexts [97]. The attacker may start by gathering ciphertexts and their corresponding plaintexts, if available, and then use these pairs to derive information about the key, either through direct computation or by leveraging patterns and statistical anomalies.

The impact of a successful key recovery attack is significant and far-reaching. Once the secret decryption key is compromised, the attacker can decrypt all past and future communications encrypted with that key, leading to a total breach of confidentiality. This can result in the exposure of sensitive information, financial loss, privacy violations, and damage to the reputation of the affected entities. In the context of secure communication systems, key recovery attacks can undermine trust and render the encryption scheme useless, necessitating immediate corrective actions, such as key replacement, algorithm updates, and enhanced security measures to prevent future attacks. Cheng's work [76] presents a key-recovery attack that utilises the timing leakage of the side channel attack, which can determine the secret key with a few ciphertexts. In the work of Guo [98], key recovery attacks on Approximate Homomorphic Encryption, such as CKKS are performed. The attack is performed on OpenFHE, demonstrating the ability to recover the secret key using just one shared decryption output.

Researchers have revealed that several SHE schemes are vulnerable to key recovery attacks, stronger than typical IND-CCA1 attacks, allowing an adversary to recover private keys through decryption oracle queries [99]. Adaptive key recovery attacks were carried out on NTRU-based SHE schemes, enabling the private key to be computed for all parameter sets. These attacks demonstrate the need for extreme caution when deploying HE, as the presence of such attacks indicates that the scheme is not CCA1-secure. Indeed, nearly every SHE scheme proposed thus far has been susceptible to this type of attack.

4.2.1 Defense mechanism 1: noise flooding

Noise-flooding techniques are different from noise injection. Noise injection adds random noise to specific parts of the computation process, such as during intermediate calculations or data transfers. On the other hand, noise flooding is more aggressive and may affect the entire operation or larger portions of the system. The goal is to overwhelm the whole system with noise to make it more difficult for an attacker to distinguish the useful signal from the noise. Noise-flooding techniques have been proposed as an effective countermeasure against IND-CPAD attacks [100]. These methods involve adding noise (from a Gaussian distribution) to the message obtained after decrypting a ciphertext before returning it to the adversary. Li, Micciancio, Schultz, and Sorrell formally analysed these techniques, showing that when the noise level is sufficiently high, they can be provably secure. However, the noise required for this level of security is substantial, significantly limiting the message precision CKKS can handle, typically to 8 or 16 bits for practical parameter sets [101]. In Guo's work [98], they employ noise-flooding countermeasures based on non-worst-case noise estimation, which are implemented to attain IND-CPAD security.

4.2.2 Defense mechanism 2: key escrow and split key schemes

The implementation of key escrow and split key schemes in HE systems presents unique challenges due to the complex nature of the keys involved. HE often relies on large, structured keys that enable computational operations on

encrypted data. Dividing these keys while maintaining their homomorphic properties requires careful mathematical consideration. Recent research has explored adapting traditional secret sharing schemes for use with HE keys, proposing a novel framework that preserves the structural integrity of the keys while providing the security benefits of key splitting [102].

4.2.3 Defense mechanism 3: hardware security modules (HSMs)

HSMs are designed to provide a secure environment for cryptographic operations, offering robust protection for encryption keys throughout their lifecycle - from generation and storage to usage and destruction. The primary strength of HSMs lies in their physical security measures. These devices are engineered to resist various forms of tampering, including physical intrusion, side-channel attacks, and environmental manipulations. Many HSMs are certified under stringent security standards such as FIPS 140-2 or Common Criteria, ensuring they meet rigorous security requirements. In the context of HE, keys are typically large and structured, requiring careful management to maintain their integrity and security [103]. HSMs provide a secure execution environment where these keys can be generated and stored, ensuring they never leave the protected confines of the hardware [104].

4.2.4 Defense mechanism 4: post-quantum cryptography

Post-quantum cryptographic algorithms are resistant to attack by quantum computers. It can provide long-term security for encryption keys in the case of safeguarding against future advancements in computational power. Post-quantum cryptography focuses on developing encryption algorithms based on problems that are believed to be hard for quantum computers to solve, such as lattice-based cryptography and code-based cryptography. Many HE schemes rely on lattice-based problems due to the post-quantum security it provides. The LWE problem and the Ring-LWE form the basis for several HE schemes that are believed to be resistant to quantum attacks. Ongoing research is focused on optimizing these schemes for practical implementation while maintaining their quantum resistance [105]. On the other hand, code-based cryptography is less common in the current HE systems. It offers another avenue for post-quantum security. Researchers are exploring ways to develop homomorphic schemes based on error-correcting codes, which could provide an alternative approach to quantum-resistant HE [106].

4.3 Chosen ciphertext attack (CCA) and defenses

A CCA is a scenario where an adversary gains the capability to choose ciphertexts and obtain their corresponding plaintexts, not directly but via access to a decryption oracle. The attacker can submit any ciphertext of their choosing to this oracle and receive the plaintext output. The critical aspect of a CCA is that the attacker uses this information to deduce or break the underlying encryption scheme, potentially recovering the secret key or finding a way to decrypt other ciphertexts without direct access to the key [107].

In more technical terms, there are two main types of CCA. The first is CCA1 (Non-Adaptive Chosen Ciphertext Attack). Here, the attacker chooses ciphertexts to be decrypted before obtaining any information about the target ciphertext. This form is considered less powerful [108]. The second type is CCA2 (Adaptive Chosen Ciphertext Attack). The more formidable variant is where the attacker can adaptively choose subsequent ciphertexts based on information gathered from previous decryptions. This adaptability significantly enhances the attacker's ability to exploit the encryption scheme [109]. In general encryption algorithms, CCAs represent a significant vulnerability. If an encryption scheme is not designed to be CCA-secure, even ciphertexts that seem robust under other attack models might be vulnerable to decryption or key recovery by a determined adversary. CCA security is thus a critical benchmark in evaluating the robustness of an encryption algorithm. For instance, in standard encryption systems like RSA or ElGamal, CCA attacks can be devastating if the schemes are implemented without safeguards like padding schemes (e.g., Optimal Asymmetric Encryption Padding or OAEP) designed specifically to counteract these attacks [110].

HE presents a unique challenge when considering CCAs. HE allows computation on encrypted data without decrypting it, preserving the confidentiality of the underlying data. This property is immensely valuable in scenarios like secure data analysis, cloud computing, and privacy-preserving ML. However, the same property that makes HE powerful also makes it particularly vulnerable to CCAs. In a HE scheme, an attacker could, theoretically, craft specific ciphertexts that, when decrypted, provide insights into the secret key or the plaintext data. The adaptive nature of CCA2 is especially

problematic; since HE involves multiple stages of computation on ciphertexts, each stage could inadvertently leak information if the scheme is not rigorously CCA-secure [111].

4.3.1 Defense mechanism 1: CCA-secure homomorphic encryption schemes

Developing HE schemes that are secure against CCAs has been a key research focus in recent years. Table 3 presents the standard definitions of security nowadays. Traditional CCA-secure encryption schemes often conflict with the malleable nature of HE. Several approaches have been proposed to address this challenge. Constrained HE limits the homomorphic operations on ciphertexts, reducing the attack surface while allowing some computational capabilities [31]. Randomisation Techniques transform ciphertexts to preserve the plaintext but alter the ciphertext's structure, preventing attacks from chosen ciphertext queries [112]. Hybrid Schemes combine HE with other cryptographic primitives, such as zero-knowledge proofs or digital signatures, to provide CCA security. Functional Encryption Approaches use functional encryption techniques to achieve CCA security by allowing fine-grained control over computations on encrypted data [113].

4.3.2 Defense mechanism 2: ciphertext randomization

Randomising the ciphertext is an effective defense mechanism against CCA. This process involves introducing random elements, such as padding or randomised encryption schemes, into the encryption process. As a result, even if the same plaintext is encrypted multiple times, the resulting ciphertext will be different each time. This randomness makes it much harder for attackers to identify patterns or perform statistical analysis on the ciphertext. Randomisation ensures that attackers cannot easily infer relationships between ciphertexts, even if they have knowledge of some plaintext-ciphertext pairs [114]. By disrupting the predictability of the encryption process, this method effectively counters CCA.

4.3.3 Defense mechanism 3: ciphertext integrity and authentication

The homomorphic signatures allow for the signing of data in such a way that computations can be performed on the signatures, mirroring the operations on the underlying ciphertexts. This enables verification of both the integrity of the ciphertext and the correctness of homomorphic computations. Verifiable Computation allows a client to outsource computations to a server and efficiently verify the correctness of the returned results [115]. In the context of HE, verifiable computation schemes can be used to ensure that homomorphic operations have been performed correctly, mitigating risks associated with malicious modifications of ciphertexts. Authenticated HE combines HE with authentication mechanisms, allowing for both computation on encrypted data and verification of the ciphertext's integrity. Recent research has proposed various constructions that achieve both homomorphic properties and ciphertext authenticity. Homomorphic Message Authentication Codes are MAC schemes that support homomorphic operations, allowing for the verification of both the integrity of individual ciphertexts and the correctness of computations performed on them [116].

4.3.4 Defense mechanism 4: regular audits and updates

Regularly auditing cryptographic implementations and updating encryption algorithms and protocols can help identify and mitigate vulnerabilities that could be exploited in ciphertext-only attacks. Audits involve thorough reviews of cryptographic systems, including code analysis, penetration testing, and compliance checks. Regular updates ensure that the latest security patches and improvements are applied to the encryption system. Implementing strict access controls on who can submit ciphertexts for decryption or homomorphic computation can limit the opportunities for CCAs. Comprehensive audit logging of all operations can also help detect and respond to potential attacks [117].

4.4 Known plaintext attacks (KPA) and defenses

In KPA, the attacker has access to both the plaintext and its corresponding ciphertext. The goal is to use this information to deduce the encryption key or gain insights into the encryption process. KPAs can be particularly effective if the attacker has access to a large number of plaintext-ciphertext pairs. KPA is particularly potent against encryption schemes that do not use additional randomisation, such as static initialisation vectors or lack of proper padding. The primary challenge in a KPA is obtaining the initial set of plaintext-ciphertext pairs, but once acquired, the attack can significantly compromise the encryption system [118].

Table 3 Summary of security notions in cryptography

Security Notion	Full name	Usage	Protects against	Key concepts
IND-CPAD	Indistinguishability under Chosen Plaintext and Divisibility Attack	Used in scenarios where an attacker might have partial information about the plaintext	CPA and attacks exploiting divisibility properties of the ciphertext	Attacker can't distinguish between encryptions of chosen plaintexts. Ciphertext doesn't reveal divisibility properties of plaintext
IND-CPA	Indistinguishability under Chosen Plaintext Attack	Standard security notion for public-key encryption	CPA	Attacker can encrypt arbitrary messages. Attacker can't distinguish which of two chosen plaintexts was encrypted
IND-CCA	Indistinguishability under Chosen Ciphertext Attack	Stronger notion used in scenarios where an attacker might have temporary access to decryption oracle	CCAs (non-adaptive)	Attacker can decrypt chosen ciphertexts (except the challenge ciphertext). Attacker still cannot distinguish which of two chosen plaintexts was encrypted
IND-CCA2	Indistinguishability under Adaptive Chosen Ciphertext Attack	Strongest commonly used security notion for public-key encryption	Adaptive CCAs	Attacker can decrypt chosen ciphertexts even after seeing the challenge ciphertext. Provides security against a wide range of practical attacks

Two common methods for using plaintext and ciphertext to identify encryption keys are frequency analysis and pattern matching. In frequency analysis, the attacker uses the frequency of letters or symbols in the ciphertext to determine the encryption key. For example, if the attacker knows that a particular letter appears very frequently in the plaintext, they can infer that it corresponds to a common letter in the ciphertext, such as “e” or “a”. By comparing the frequency of letters in the ciphertext to the frequency of letters in the plaintext, the attacker can deduce the encryption key. In pattern matching, the attacker looks for patterns in the plaintext and compares them to patterns in the ciphertext. For example, if the attacker knows that a particular phrase appears frequently in the plaintext, they can look for that phrase in the ciphertext and use the corresponding encryption key to decrypt the message [119].

If the plaintext and ciphertext are identical, the attacker can use the pattern-matching method to deduce the encryption key. In this case, the attacker can identify the encryption key by comparing the patterns in the plaintext and ciphertext. However, if the plaintext and ciphertext are not identical, the attacker must use frequency analysis to deduce the encryption key. The process of performing a KPA involves analysing the known plaintext and its corresponding ciphertext to identify patterns or relationships that can reveal the key. For instance, in classical ciphers, an attacker might look for repeating patterns in the ciphertext that correspond to known phrases in the plaintext. In modern cryptographic systems, an attacker might use differential cryptanalysis or linear cryptanalysis techniques to uncover the key. The impact of a successful KPA can be severe, as it may allow the attacker to decrypt any future ciphertexts encrypted with the same key [120].

Researchers perform KPA on Omar and abed HE Scheme, which is a noise-free FHE scheme that allows arbitrary computations on encrypted data without decryption [121]. KPA towards HE scheme for cloud computing, resulting in 4 of the proposed homomorphic schemes is vulnerable to KPA [122]. KPA on polynomial-based HE, resulting in some cryptosystems being insecure against KPA if there are at least two pairs [123].

4.4.1 Defense mechanism 1: key management and randomization techniques

Dynamic key generation, key rotation policies, randomised encryption, and entropy injection are critical techniques for enhancing the security of HE systems, particularly in mitigating KPAs. Dynamic key generation ensures that unique encryption keys are created for each session or message, reducing the effectiveness of attacks by limiting exposure to static keys. Key rotation policies further protect encrypted data by periodically updating encryption keys, minimising the time frame within which an attacker could exploit plaintext-ciphertext pairs. Randomised encryption introduces randomness into the encryption process, ensuring that the same plaintext generates different ciphertexts each time it is encrypted, thereby preventing pattern recognition. Additionally, entropy injection regularly adds unpredictability to the key generation and encryption processes, strengthening the overall security and resilience of the HE system against potential cryptanalytic attacks [124].

4.4.2 Defense mechanism 2: strong cryptographic protocols

Employing cryptographic protocols that are resistant to KPAs, such as those using strong encryption algorithms and key management practices, can protect against these types of attacks. These protocols are designed to ensure that even if an attacker knows the plaintext and ciphertext, they cannot deduce the encryption key or gain meaningful insights into the encryption process. Researchers proposed a Fully symmetric HE that is immune to known-plaintext attack and known ciphertext attack [125]. The new scheme is referred to as the Homomorphic Hybrid Symmetric Encryption Scheme. It is based on mixing the homomorphic behaviour of two well-known symmetric encryption schemes.

4.4.3 Defense mechanism 3: data anonymization

Data anonymisation serves as a valuable defense mechanism against KPA. Even if an attacker has access to both the plaintext and its corresponding ciphertext, anonymised data is less useful for inferring the encryption key or other sensitive information. Anonymisation techniques focus on removing or obfuscating personally identifiable information and other critical details before the encryption process begins. This additional layer of protection makes it more difficult for attackers to link encrypted data to meaningful insights [126]. As a result, even when plaintext-ciphertext pairs are available, the utility of the data for cryptanalysis is significantly reduced.

4.4.4 Defense mechanism 4: secure software development practices

Adhering to secure software development practices, including thorough testing, code reviews, and adherence to security standards, can prevent vulnerabilities that could be exploited in KPAs. Secure software development practices involve using secure coding techniques, performing regular security assessments, and following best practices for cryptographic implementations. Following secure coding practices, using vetted cryptographic libraries, and conducting regular code reviews are essential in developing robust HE systems resistant to CCAs [127].

4.5 Chosen plaintext attacks (CPA) and defenses

A CPA is a scenario where the attacker can select plaintexts and observe their corresponding ciphertexts. While considered less practical than a KPA, it remains highly dangerous [128]. If a cipher is vulnerable to a KPA, it is automatically susceptible to a CPA, though the reverse is not always true. In modern cryptography, differential cryptanalysis is a notable example of a CPA, and it's one of the rare techniques where conversion from a CPA to a KPA is possible due to its focus on text pairs. In a CPA, the attacker can freely select arbitrary plaintexts and receive the corresponding ciphertexts. The goal is to gain information about the encryption key or the encryption algorithm. CPAs are particularly relevant in scenarios where attackers can interact with the encryption system and obtain ciphertexts for chosen plaintexts [101].

There are two types of CPAs. The first is the batch-CPA, where the attacker selects all plaintexts in advance without initially seeing any corresponding ciphertexts. This is typically what's meant by a "CPA" unless otherwise specified. The second type is the adaptive CPA (CPA2), where the attacker can request ciphertexts for additional plaintexts after observing the ciphertexts of previously chosen plaintexts [129].

To perform a CPA, the attacker first identifies the objective, such as recovering the encryption key or decrypting sensitive information, and gains an understanding of the encryption scheme in use. Next, the attacker sets up the environment by establishing interaction with the encryption oracle and selecting initial plaintexts based on prior knowledge or hypotheses. The attacker then submits these plaintexts to the oracle, collects the resulting ciphertexts, and analyses them to identify patterns or anomalies. Through iterative testing and correlation analysis, the attacker refines their hypotheses and tests them to recover the encryption key or exploit algorithmic weaknesses. The success of the attack is evaluated based on key recovery accuracy, efficiency, and the extent of the breach, followed by comprehensive documentation of the methodology, findings, and security recommendations. Practical considerations include ensuring adequate computational resources and adhering to ethical and legal guidelines. Understanding and mitigating CPAs is crucial for developing robust cryptographic systems capable of withstanding such sophisticated attacks [130].

On the other hand, a CPA2 is more dynamic and powerful. In a CPA2, the attacker can adaptively choose plaintexts based on the ciphertexts already observed. After analysing the initial set of plaintexts and their corresponding ciphertexts, the attacker can select additional plaintexts and obtain more ciphertexts, allowing for a more iterative and refined analysis. This adaptive nature makes CPA2 more effective in uncovering subtle weaknesses in the encryption scheme, as the attacker can continuously refine their hypotheses and exploit new information gained during the attack.

The primary difference between CPA and CPA2 lies in the adaptability of the attack process. While CPA requires all plaintexts to be chosen before any ciphertexts are seen, CPA2 allows for continuous interaction with the encryption oracle, making it more flexible and potentially more damaging. The impact of CPA2 is generally more severe because it can reveal more intricate details about the encryption scheme, leading to a higher likelihood of breaking the encryption and compromising security. Both types of attacks highlight the importance of designing encryption protocols that are resilient against such sophisticated attacks.

4.5.1 Defense mechanism 1: randomized encryption

Randomised encryption schemes, which incorporate random elements into the encryption process, ensure that the same plaintext will produce different ciphertexts each time it is encrypted. This prevents attackers from gaining useful information by comparing ciphertexts. Techniques like probabilistic encryption and adding random padding to plaintexts before encryption can be used to introduce randomness into the ciphertexts. In 1999, Paillier introduced a probabilistic encryption scheme, with its security relying on the computational difficulty of the decisional composite residuosity

assumption. Encryption using the Paillier algorithm offers semantic security, protecting against chosen-plaintext attacks (IND-CPA) [38].

4.5.2 Defense mechanism 2: use of strong cryptographic primitives

One of the primary defenses against CPA is the use of strong cryptographic primitives. These are mathematical algorithms specifically designed to resist attacks, including CPA. Cryptographic primitives based on hard mathematical problems, such as lattice-based cryptography, offer significant security advantages. Lattice-based systems, in particular, rely on problems that are difficult to solve even with the most advanced computational techniques. This makes it highly challenging for attackers to extract any meaningful information from chosen plaintexts. The encryption process in these schemes produces ciphertexts that are highly non-linear and unpredictable, effectively preventing any clear correlations between plaintexts and ciphertexts. Consequently, even when attackers can choose the plaintexts, they cannot easily derive the corresponding ciphertext patterns or reverse-engineer the encryption process [131]. These cryptographic primitives are essential for ensuring that sensitive data remains secure, even in the face of sophisticated cryptanalytic attacks.

4.5.3 Defense mechanism 3: adaptive security measures

The implementation of adaptive security measures can be used to defend against CPA. Adaptive security refers to techniques that adjust key or encryption parameters over time, reducing the likelihood of successful attacks. By frequently changing encryption keys or parameters, adaptive security measures limit the duration in which attackers can gather sufficient data to conduct a meaningful CPA. Key rotation, session keys, and dynamic re-keying are all examples of techniques that can be employed to ensure that encryption keys are regularly updated. This makes it much more difficult for attackers to accumulate enough data to mount a successful attack. Adaptive security measures essentially reduce the window of opportunity for attackers, as they prevent prolonged use of the same cryptographic key or setup [132]. As a result, these measures significantly enhance the overall security of cryptographic systems by reducing the potential for key exposure and compromising encrypted data.

4.5.4 Defense mechanism 4: secure key management

Strong key management practices are a critical aspect of defending against CPA. Poor key management can provide attackers with the opportunity to exploit encryption keys, potentially compromising the entire encryption process. Secure key management involves a variety of strategies aimed at safeguarding encryption keys from unauthorised access or misuse. These include frequent key rotation, where keys are replaced or updated regularly, and secure storage of keys, often within HSMs. HSMs are dedicated hardware devices that offer a higher level of protection for keys than software solutions alone. Additionally, implementing strict access controls and multi-factor authentication ensures that only authorised personnel can access encryption keys. By following robust key management protocols, organisations can effectively prevent attackers from using CPA to compromise the integrity of their encryption systems [133]. Proper key management serves as a foundational element in maintaining the security of encrypted data.

4.6 Fault injection attack (FIA) and defenses

FIA aims to introduce error into the HE process, either during computation or key generation, to uncover secret keys or plaintexts. FIAs can be software-based or hardware-based. Figure 3 shows the invasive attack and the non-invasive attacks in FIA. The hardware-based FIA can be conducted by manipulating the environment where cryptographic operations occur, such as through voltage manipulation, electromagnetic radiation, or even laser-based techniques [134], as shown in Table 4.

A majority of FI attacks involve either hardware tampering or the use of physical probes connected to the target circuitry. For instance, voltage glitching attacks usually require the desoldering of power filtering capacitors so that the attacker is able to have more control of the voltage input of the processor. In some instances, FI attacks can be executed through software alone, without modifying the hardware. This is possible in cases where System-on-Chips include software-controlled voltage regulators and clock sources across different power and clock zones. Two examples of such attacks are CLKSCREW and Plundervolt. These voltage regulators and clock sources are typically used

Fig. 3 Types of fault injection attack

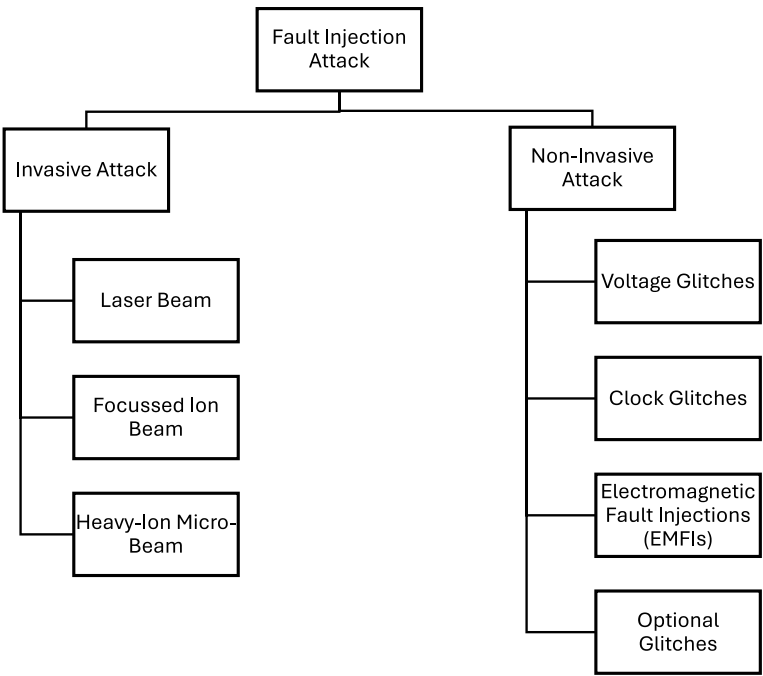


Table 4 Hardware-based FIA—types of glitches

Type of glitches	Description	References
Voltage	Momentary voltage drops during operations to cause faults	[135]
Clock	Modifying clock signals to disrupt timing constraints with glitch pulses	[135]
EM	High-intensity EM pulses induce currents in chip circuits	[136, 137]
Optical	Uses infrared lasers on exposed silicon die to cause faults	[138]

for Dynamic Voltage and Frequency Scaling to conserve power. Nonetheless, access to these controls should not be accessible to low-privilege software, and safeguards must be in place to stop malicious software from causing faults.

There are three main types of faults in FIA: instruction skipping, incorrect data fetch, and incorrect instruction fetch or decode. Instruction skipping ignores the execution of the currently processed instruction. Powerful attacks can be introduced by using this fault model, such as privilege escalation, a simple key extraction, or a neural network misclassification [135]. It can happen with or without repeating the previous instruction. While the problem of skipping a single instruction is relatively easy to induce, skipping multiple nearby instructions is more challenging [139]. Incorrect data fetch results in uncontrolled bit flips within a read word, or the entire word is read as either all ones or all zeros. This might cause the software to revert to a less secure state since security-critical data is corrupted in this way. Bit-reset faults change bits from one to zero, while bit-set faults flip bits from zero to one. It’s easier to induce single-bit flips than multiple-bit flips, although controlling which specific bit flips is difficult. Changing a read word to a specific value, aside from all ones or zeros, is also difficult to achieve through glitching [140]. Incorrect instruction fetch or decode causes the later pipeline stages to evaluate the wrong instruction. Instructions that consist mostly of ones or zeros are easier to trigger via glitching than specific instructions with a mix of ones and zeros. It is challenging to alter a fetched or decoded instruction to a specific arbitrary value due to the difficulty of controlling which bits are flipped by the glitch [141].

The authors proposed a universal solution for efficiently detecting prominent FIAs [141]. The proposed sensor functions by translating the consequences of fault attacks into measurable “time” differentials. When an FIA is performed, the altered data will often fail to pass the error detection check, triggering an error-handling routine. This mechanism ensures that any injected faults are recognised before they can cause further damage, such as compromising sensitive information or leading to incorrect system behaviour. However, while error detection codes are effective, they may not prevent the fault but can help in detecting and mitigating its effects, thus limiting the attack’s success [142].

Table 5 Lattice-based homomorphic encryption: key aspects and challenges

Aspect	Description
Target	Private key recovery or ciphertext decryption in lattice-based HE schemes
Key algorithms	LLL, BKZ
Main challenges	SVP, CVP
Complexity	Exponential in lattice dimension for exact solutions
Key factors	Lattice dimension, basis quality, HE scheme parameters
Common targets	LWE-based schemes, Ring-LWE-based schemes
Defense strategies	Increasing lattice dimension, careful parameter selection, noise management
Trade-offs	Security vs. performance and ciphertext size

4.6.1 Defense mechanism 1: physical inaccessibility

In previous work, the authors [143] approach is to make the system's implementation physically inaccessible to potential attackers. By placing critical components in secure, tamper-resistant environments, such as sealed enclosures or hardened chips, it becomes significantly more difficult for an adversary to physically access the hardware necessary to inject faults. This method relies on the principle of security through obscurity, where the less accessible the hardware, the lower the chance of successful tampering. For example, systems can be embedded in devices that are difficult to disassemble without destroying the hardware, or they can be enclosed in tamper-evident packaging. Furthermore, shielding techniques can be used to protect against certain types of electromagnetic or optical fault injection methods. While physical inaccessibility does not provide absolute security, it significantly raises the bar for attackers, making FIAs less feasible.

4.6.2 Defense mechanism 2: use of nontrivial constants

Using nontrivial constants in cryptographic algorithms and other sensitive operations can be an effective countermeasure against FIAs. Fault injections often rely on predictable or trivial constants that, when altered, produce predictable results. By using nontrivial, random, or dynamically generated constants, the effect of a fault injection becomes much harder to predict or exploit [143]. For example, in cryptographic algorithms, employing nontrivial constants in a key generation or in the encryption process can thwart fault injection attempts that aim to deduce the key or plaintext. This approach increases the complexity of the attack, as the attacker cannot easily determine the effect of the injected fault, thus reducing the likelihood of successfully compromising the system.

4.6.3 Defense mechanism 3: fault injection-resistant Implementations

A proactive defense mechanism is to search for and adopt implementations that are inherently resistant to FIAs. This involves designing hardware and software in such a way that they are less susceptible to faults, whether intentional or accidental. Fault injection-resistant implementations can be achieved through rigorous testing, formal verification methods, and the use of fault-tolerant design principles [144]. For instance, redundancy in processing, where the same operation is performed multiple times with the results compared for consistency, can help detect and correct faults. Similarly, designing algorithms with inherent resistance to faults—such as those that do not produce exploitable side effects when faults occur—can be a powerful defense. Regular audits and updates to these implementations ensure they remain resistant as new attack techniques emerge.

4.7 Lattice attack and defenses

Lattice attacks represent a significant threat to many HE schemes, particularly those based on lattice problems such as LWE and Ring-LWE. These attacks exploit the mathematical structure of lattices to attempt to recover the private key or decrypt ciphertexts. The fundamental approach in lattice attacks involves solving the Shortest Vector Problem (SVP) or the Closest Vector Problem (CVP) in high-dimensional lattices. Attackers typically employ lattice reduction algorithms, such as the Lenstra-Lenstra-Lovász (LLL) algorithm or the Block Korkine-Zolotarev (BKZ) algorithm, to find short vectors in the lattice. The effectiveness of these attacks depends on various factors, including the dimension of the lattice, the quality of the lattice basis, and the specific parameters of the HE scheme. As HE schemes often rely on the hardness of

Table 6 Types of attacks on homomorphic encryption

Attack type	Description	References
Short vector attacks	These attacks attempt to find the shortest vector in a lattice, which corresponds to the private key or a related secret in the cryptographic scheme. This is the foundation of the SVP	[145, 146]
LWE attacks	These attacks focus on solving the LWE problem, which underpins many HE schemes. The LWE problem involves a system of noisy linear equations, and the attacker's goal is to recover the original variables or the noise	[147, 148]
Decoding attacks	These attacks involve solving the CVP, where the attacker tries to find a lattice point closest to a given target point (which could represent ciphertext or intermediate encrypted data)	[149]
Hybrid attacks	Hybrid attacks combine lattice reduction techniques with brute-force search over some smaller parts of the key	[150]

Table 7 Lattice attacks on homomorphic encryption: concepts and processes

Category	Concept	Description
Foundational problems	LWE [151]	The LWE involves recovering a secret vector s from samples $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + e)$ where a is a random vector and <i>e</i> is a small error term. This problem forms the basis for numerous lattice-based cryptographic schemes.
	Ring-LWE [152]	Ring-LWE operates over polynomial rings, providing enhanced efficiency compared to standard LWE. Samples are of the form $(a, b = a \cdot s + e)$ where <i>a</i> , <i>s</i> , and <i>e</i> are polynomials.
Lattice reduction algorithms	LLL [153]	The Lenstra-Lenstra-Lovász algorithm is a polynomial-time method producing basis vectors of moderate quality. Output vectors are at most $2^{n/4}$ times longer than the shortest vector.
	BKZ [154]	Block-Korkine-Zolotarev is a more powerful generalisation of LLL operating on blocks of vectors balancing running time and output quality.
Core lattice problems	SVP [155]	The Shortest Vector Problem aims to find a non-zero vector with minimal Euclidean norm in a lattice. Its complexity grows exponentially with the lattice dimension. Approximate solutions often suffice for attacks.
	CVP [156]	The Closest Vector Problem involves finding the lattice vector nearest to a given target vector. It is utilised for ciphertext decryption in certain schemes and is generally considered at least as hard as SVP.
Lattice attack process	1. Lattice construction	Build lattice based on public parameters of HE scheme.
	2. Lattice reduction	Apply LLL or BKZ to find short vectors.
	3. Information extraction	Exploit short vector structure to recover key or decrypt.

lattice problems for their security, understanding and mitigating lattice attacks is crucial for ensuring the robustness of these cryptographic systems.

Table 5 provides a concise overview of the key aspects of lattice attacks on HE, highlighting the main algorithms, challenges, and factors that influence their effectiveness. It also touches on common defense strategies and the inherent trade-offs in securing HE schemes against these attacks. Table 6 summarizes four types of lattice attacks on Homomorphic encryption, while Table 7 categorizes three lattice problems and explains the underlying concepts and logic.

4.7.1 Defense mechanism 1: increasing lattice dimension

In lattice-based cryptography, one of the most effective methods of defending against lattice attacks is to increase the dimension of the lattice. A higher lattice dimension exponentially increases the difficulty for attackers in carrying out a successful attack using current lattice reduction techniques. For example, in HE schemes, increasing the size of the lattice requires larger keys and ciphertexts. This increase in size, however, makes it much more difficult for an attacker to break the encryption or solve the underlying mathematical problems, such as the Shortest Vector Problem (SVP). As demonstrated in Chen's work, the experimental values of Δ , which represent successful attack instances, decrease as the lattice dimension increases. This indicates that larger lattices provide stronger security because they require significantly more computational resources to attack effectively.

4.7.2 Defense mechanism 2: noise management

Proper noise management is another critical defense mechanism in lattice-based cryptographic schemes. The security of these schemes relies heavily on the hardness of the LWE problem, which is closely related to the amount of noise introduced during the encryption process. By adding sufficient noise, the complexity of the underlying mathematical problem is maintained, making it much more resistant to lattice attacks. However, managing this noise must be done carefully. Too little noise can make the encryption vulnerable to attacks, while too much noise can degrade the performance or render the ciphertext unusable. Properly balanced noise ensures that the encryption remains secure without sacrificing efficiency. Noise management is an essential component in preserving the difficulty of lattice-based problems, thereby protecting against attacks that attempt to exploit weaknesses in the encryption [157]. This technique helps to maintain the robustness of cryptographic schemes against a wide range of attacks.

4.7.3 Defense mechanism 3: secure parameter selection

Choose cryptographic parameters (e.g., key size, noise level) carefully to ensure they are resistant to known attacks. Cryptographic libraries and standards like NIST's post-quantum cryptography project offer guidelines for secure parameter selection [158]. Lattice attacks pose a serious threat to HE schemes, especially those based on lattice problems like LWE and Ring-LWE. These attacks exploit the mathematical structure of lattices to recover private keys or plaintext data, often through lattice reduction algorithms such as LLL or BKZ. To defend against lattice attacks, cryptographic schemes must use high-dimensional lattices, properly manage noise levels, and select secure parameters. By following these best practices, HE can remain secure even against sophisticated lattice-based attacks.

5 Open research directions

While HE provides a robust framework for secure data computations, ongoing research is crucial to address emerging threats and improve its efficiency. One of the most pressing challenges lies in enhancing the efficiency of HE schemes. Computational overhead and latency are limiting the development of HE's practical application. Researchers are exploring various techniques to mitigate these limitations, including optimised algorithms, hardware acceleration, and parallel processing. These efforts aim to make HE systems more feasible for real-world scenarios by reducing the computational burden associated with encrypted data processing.

Another critical area of research is post-quantum security. Cybersecurity quantum attacks assume quantum computers are used to attack traditional cryptographic schemes. Traditional cryptographic methods may be vulnerable to quantum attacks, including the current HE system. As a result, researchers are focusing on developing post-quantum HE schemes

that can withstand future quantum threats. Ensuring that HE remains secure in the post-quantum era is essential to protect encrypted data in the long term.

Hybrid cryptographic approaches are gaining attention in addition to improving individual HE schemes. Researchers aim to enhance both security and functionality by combining HE with techniques like secure multi-party computation and differential privacy. Hybrid methods leverage the strengths of multiple cryptographic techniques to provide more comprehensive protection for sensitive data. For instance, secure multi-party computation enables secure collaborative computations between multiple parties, while differential privacy ensures that the results of data analysis do not leak individual information, even when using HE.

Another challenge is scalability, particularly for handling large datasets and complex computations. As more industries seek to implement HE in various applications, it becomes essential to develop scalable HE schemes that can efficiently manage the processing of large data volumes. Researchers are working on creating HE solutions that not only maintain security but also offer the scalability needed for widespread adoption in areas like healthcare, finance, and cloud computing.

Finally, improving the usability of HE systems is crucial for broader implementation. Many current HE systems are complex and require specialised knowledge to operate, limiting their accessibility to non-experts. Enhancing usability involves developing more intuitive user interfaces, providing clear documentation, and offering training resources to help end-users, particularly in industries where encryption expertise is limited. Making HE more user-friendly will be key to facilitating its adoption across various sectors.

In conclusion, while HE already offers significant security benefits, there is still much work to be done in terms of improving efficiency, quantum resistance, scalability, and usability. Ongoing research in these areas, as well as hybrid approaches that combine HE with other cryptographic methods, will be critical to ensuring that HE can meet the demands of future applications.

6 Conclusion

Healthcare digitises and powers advanced computational techniques for data analysis and service delivery. The importance of secure and privacy-preserving technologies like HE cannot be exaggerated. This work has explored HE in healthcare applications. The focus is on the attacks endangering these systems and the corresponding defense mechanisms. The application of HE in healthcare domains showcases its potential in secure data processing. HE has proved its ability to balance data utility with strict privacy requirements. HE has been applied to secure electronic health records, enable privacy-preserving genomic data analysis, encourage secure medical imaging, and progress privacy-preserving ML techniques. The emerging fields of secure federated learning, privacy-preserving clinical trials, and secure remote monitoring further show the role of HE in improving the future of healthcare technology.

However, the adoption of HE in healthcare is challenging. Attacks against HE systems reveal a complex threat landscape requiring robust defense strategies. Side-channel attacks, including timing, power analysis, and EM attacks, have posed considerable risks to the confidentiality of encrypted healthcare data. Key recovery attacks jeopardise the foundational security of HE systems, while CCA, KPA and CPA exploit vulnerabilities in encryption schemes to compromise data integrity and confidentiality. Examining FIA and lattice attacks further highlights the complexity of threats facing HE implementations in healthcare. These attacks exploit hardware vulnerabilities and mathematical flaws in cryptographic schemes.

Various defense mechanisms are studied to respond to the threats. Constant-time algorithms, noise injection, and power analysis countermeasures offer security against side-channel attacks. Key management techniques, including noise flooding, key escrow schemes, and HSMs, strengthen defenses against key recovery attempts. The CCA-secure HE schemes, ciphertext randomisation, and integrity verification mechanisms address vulnerabilities to CCA. For known plaintext and CPA, strategies such as strong cryptographic protocols, data anonymisation, and adaptive security measures provide essential layers of protection. Physical security measures, fault-resistant implementations, and secure parameter selection form the backbone of defenses against fault injection and lattice attacks.

In conclusion, HE is a powerful tool that provides healthcare with privacy-enhancing technologies. The ongoing development of defense mechanisms offers a promising outlook even though the challenges of various attacks are significant. By addressing current vulnerabilities and expecting future threats, the healthcare industry can harness the full potential of HE to advance patient care, research, and data-driven decision-making while steadfastly protecting individual privacy and data security.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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